1.1. $\overrightarrow{BA}$

1.2. $\vec{a} = (6; 8)$

$|\vec{a}| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = 10$

$\vec{a_0} = \frac{1}{10}(6; 8) = \left(\frac{6}{10}; \frac{8}{10}\right)$

1.3 $\vec{b} = (-2; 1; 2)$

$|\vec{b}| = \sqrt{-2^2 + 1^2 + 2^2} = \sqrt{9}$

$\vec{b_0} = \frac{1}{\sqrt{9}}(-2; 1; 2) = \left(\frac{-2}{\sqrt{9}}; \frac{1}{\sqrt{9}}; \frac{2}{9}\right)$

2.1 $\vec{a} = (2; -1; 3)$

$\vec{b} = (6; -3; 9)$

$\lambda = 3$

2.2. $\vec{a} = (1; 2; -1)$

$\vec{b} = (-2; -4; 2)$

$\lambda = -2$

2.3.

$(2, x, 6)$

$(1, -2, 3)$

$\lambda = 2$

$x = -2 \cdot 2 = -4$

$\vec{a} = (3, -2, 1) \quad \vec{b} = (-1, 4, 2)$

5.1 $\vec{a} + \vec{b} = (3 + (-1), -2 + 4, 1 + 2)$

$\vec{a} + \vec{b} = (2, 2, 3)$

5.2. $\vec{a} - \vec{b} = (3 - (-1), -2 - 4, 1 - 2)$

$\vec{a} - \vec{b} = (4, -6, -1)$

5.3. $2\vec{a} - 3\vec{b}$

$2\vec{a} = 2(3, -2, 1) = (6, -4, 2)$

$3\vec{b} = 3(-1, 4, 2) = (-3, 12, 6)$

$2\vec{a} - 3\vec{b} = (3 - (-3), 4 - 12, 2 - 6) = (6, -8, -4)$

5.4. $\vec{a} = (1, 2, -4) \quad \vec{b} = (0, -3, 2)$

$\vec{c} = (5, 1, -1)$

$3\vec{a} + \vec{b} - 2\vec{c}$

$3\vec{a} = 3(1, 2, -4) = (3, 6, -12)$

$2\vec{c} = 2(5, 1, -1) = (10, 2, -2)$

$3\vec{a} + \vec{b} - 2\vec{c} = (3 + 0 - 10, 6 + (-3) - 2,$
$-12 + 2 - (-2) = (-7, 1, -8)$

6.1. $\vec{a} = (2, 1, -1)$ $\vec{b} = (-1, 3, 2)$

$3\vec{a} + 2\vec{b}$

$3\vec{a} = (6, 3, -3)$ $2\vec{b} = (-2, 6, 4)$

$3\vec{a} + 2\vec{b} = (6 + (-2), 3 + 6, -3 + 4) =$

$(4, 9, 1)$

6.2. $\vec{c} = (5, 1)$

$\vec{c} = 3(1, 1) + 1(2, -1)$

7.1. $\vec{a} = (1, 2)$ $\vec{b} = (3, 6)$

$3\vec{a} = \vec{b}$

$3\vec{a} - \vec{b} = 0$

$\vec{a}, \vec{b}$ линейно зависимы

7.2 $\vec{a} = (1,0)$ $\vec{b} = (0,1)$

векторы свободные

7.3. $\vec{a} = (1,0,2)$

$\vec{b}$ $(0,1,-1)$

$\vec{c} = (2,1,3)$

$2\vec{a} - \vec{c} - \vec{b} = 0$

векторы линейно зависимы

8.1. $\vec{a} = (-2,4,7)$

$\vec{a} = -2\vec{i} + 4\vec{j} + 7\vec{k}$

8.2. $\vec{e_1} = (1,0)$ $\vec{e_2} = (1,1)$

$\vec{x} = 5\vec{e_1} + 2\vec{e_2}$

9.1. $A(-2,4,1)$ $B(3,-1,5)$

$\vec{AB} = (3-(-2), -1-4, 5-1)$

$\vec{AB} = (5, -5, 4)$

$\overrightarrow{BA} = (-2-3,\ 4-(-1),\ 1-5)$

$BA = (-5,\ 5,\ 4)$

9.2. $M(1, 2, -3)$ $N(-4, 6, 0)$

$\overrightarrow{MN} = (-4-1,\ 6-2,\ 0-3)$

$\overrightarrow{MN} = -5,\ 4,\ -3$

10.1. $\vec{a} = (4, -4, 7)$

$|\vec{a}| = \sqrt{4^2 + -4^2 + 7^2} = \sqrt{16 + 16 + 49}$

$|\vec{a}| = 9$

10.2. $A(1, -2, 3)$ $B(5, 1, -1)$

$\overrightarrow{AB} = (5-1,\ 1-(-2),\ -1-3)$

$\overrightarrow{AB} = (4, 3, -4)$

$|\overrightarrow{AB}| = \sqrt{4^2 + 3^2 + -4^2} =$

$\sqrt{16 + 9 + 16} = \sqrt{41}$

11.1

$|\vec{a}| = 8 \quad \varphi = 60^\circ$

$pr_1 = 8 \cdot \frac{1}{2} = 4$

11.2

$|\vec{b}| = 12 \quad \varphi = 120°$

т.к $\varphi = 120 \Rightarrow -\frac{1}{2}$

$12 \cdot -\frac{1}{2} = -6$

11.3 я не знаю как определять углы в числе

13.1. $\vec{a} = (1,2,2)$

$a_x = 1$

$a_y = 2$

$a_z = 2$

$|\vec{a}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1+4+4}$

$|\vec{a}| = 3$

$\cos\alpha = \frac{1}{3}; \cos\beta = \frac{2}{3}; \cos\gamma = \frac{2}{3}$

$\frac{1}{3}^2 + \frac{2}{3}^2 + \frac{2}{3}^2 = \frac{1+4+4}{3} = \frac{9}{3} = 3$

13.2. $\vec{b} = (-2, 3, 6)$

$|\vec{b}| = \sqrt{-2^2 + 3^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$

$\vec{b}_x = -2$

$b_y = 3$

$b_z = 6$

$\cos_x = \frac{-2}{7}$

$\cos_y = \frac{3}{7}$

$\cos_z = \frac{6}{7}$

$\frac{-2^2}{7} + \frac{3^2}{7} + \frac{6^2}{7} = \frac{4}{7} + \frac{9}{7} + \frac{36}{7} =$

$\frac{49}{7} = 7$

16.1 $\vec{a} = (1, 3, -2)$ $\vec{b} = (4, -1, 2$

$\vec{a}\vec{b} = 1 \cdot 4 + 3 \cdot (-1) + (-2) \cdot 2 =$

$5 + (-3) + (-4) = -2$

16.2. $\vec{a} = (1, 2, -1)$ $\vec{b} = (3, -1, 1)$

$\vec{a}\,\vec{b} = 1 \cdot 3 + 2 \cdot (-1) + (-1) \cdot 1 =$

$3 + (-2) + (-1) = 0$

векторы перпендикулярны

16.3 $\vec{a} = (2, x, 1)$ $\vec{b} = (3, -2, 4)$

$\vec{a} \cdot \vec{b} = 6 + x \cdot (-2) + 4$

$x = 5$, чтобы векторы были перпендикулярны

18.1. $A(-1, 1)$, $B(2, 3)$, $C(5, -2)$

$\overrightarrow{BA} = (-1 - 2,\ 1 - 3) = (-3, 2)$

$\overrightarrow{BC} = (5 - 2,\ -2 - 3) = (3, -5)$

$|\overrightarrow{BA}| = \sqrt{-3^2 + 2^2} = \sqrt{9 + 4}$

$|\overrightarrow{BA}| = \sqrt{13}$

$|\overrightarrow{BC}| = \sqrt{3^2 + -5^2} = \sqrt{34}$

$\vec{BA} \cdot \vec{BC} = -3 \cdot 3 + 2 \cdot -5 =$
$-9 + (-10) = -19$

$\cos \angle ABC = \frac{-19}{\sqrt{13} \cdot \sqrt{34}}$

19. $pr_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

$\vec{a} = (3, 2, -1) \quad \vec{b} = (1, -2, 2)$

$|\vec{b}| = \sqrt{1^2 + (-2^2) + 2^2} = \sqrt{9} = 3$

$\vec{a}\vec{b} = 3 \cdot 1 + 2 \cdot (-2) + (-1) \cdot 2 =$
$3 - 4 - 2 = -3$

$pr_{\vec{b}} \vec{a} = \frac{-3}{3} = -1$

21.1 $\vec{a} = (2, -1, 3) \quad \vec{b} = (1, 4, -2)$
$= ((-1 \cdot -2) - (3 \cdot 4), (3 \cdot 1) - (2 \cdot -2),$
$(2 \cdot 4) - (-1 \cdot 1) = (-10, 7, 9)$

$\vec{a}\,\vec{b} = 2 \cdot 1 + (-1) \cdot 4 + 3(-2) =$

$2 + (-4) + (-6) = -8$